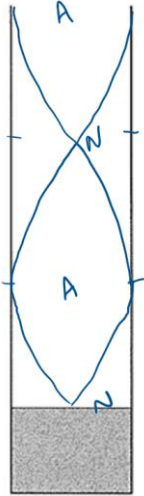


4a	Change in depth of water.	E	B1
4b	$L = \lambda/4$ $v = f\lambda$ So, $v = f(4L)$ $= 480 \times 4 \times 0.18$ $= 345.6 \text{ m s}^{-1}$ $= 346 (350) \text{ m s}^{-1}$	A	C1 A1
4ci	 A node at the water surface and an antinode at the open end of the air column Length of air column equals $\frac{3}{4}$ of wavelength of the stationary wave, with total of 2 nodes and 2 antinodes labelled correctly Correct sinusoidal shape (only award when both M marks are scored)	A	M1 M1 A1
4cii	Vibrate along axis of the tube, Vibrate with maximum amplitude.	A	B1 B1
4ciiii	$L = \frac{3}{4} \lambda$ so $\lambda = \frac{4}{3} L = \lambda'/3$ where λ' is the wavelength in part (b) Since $v = f\lambda$, $f = 3 f'$ where f' is the frequency in part (b) $f = 3 \times 480 = 1440 \text{ Hz}$	D	C1 A1